

Practice Exercises

1 Redirection & Pipes

- Use > to redirect the output of ls to files.txt.
- Append system uptime to files.txt using >>.
- Redirect standard error from ls /invalid_path to errors.txt.
- Use /dev/null to discard error messages.
- Pipe ls /etc into grep to filter files ending in .conf.

2 Working with vi and vim

- Open notes.txt in vi. Enter insert mode and type a sentence.
- Save and exit using the proper vi command.
- Try out vimtutor

3 Shell Variables & Environment

- Create a variable myname with your name as its value.
- Display the variable's value using echo.
- Convert myname to an environment variable.
- Display the contents of the PATH variable.
- Unset the myname variable and check if it still exists.

4 User & Group Management

- Create a user student_test with home directory /home/student_test.
- Set a password for student_test.
- Create a group testers and add student_test to it.
- Change student_test's primary group to testers.
- Lock student_test's account, then unlock it.

Week 3 practical/lab 2 – Managing users and groups

Administrative privileges are required.

Step 1: Create two groups

```
sudo groupadd developers
sudo groupadd testers
```

Verify:

```
getent group developers
getent group testers
```

Step 2: Create two users

```
sudo useradd -c "Developer Account" alice
sudo useradd -c "Testing Account" bob
```

Set passwords:

```
sudo passwd alice
sudo passwd bob
```

Step 3: Add users to supplementary groups

```
sudo usermod -aG developers alice
sudo usermod -aG testers bob
```

Add Alice to both groups:

```
sudo usermod -aG testers alice
```

Verify:

```
id alice
```

```
id bob
```

Expected result:

- Alice belongs to **developers** and **testers**.
- Bob belongs to **testers**.

Step 4: Check account records

```
grep -E '^(alice|bob):' /etc/passwd
```

```
getent group developers
```

```
getent group testers
```

Step 5: Set password ageing

```
sudo chage -M 90 -m 1 -W 7 alice
```

```
sudo chage -l alice
```

Expected result:

- Minimum age: 1 day
- Maximum age: 90 days
- Warning period: 7 days

Step 6: Lock and unlock an account

```
sudo usermod -L bob
```

```
sudo passwd -S bob
```

Unlock it:

```
sudo usermod -U bob
```

```
sudo passwd -S bob
```

Step 7: Clean up

```
sudo userdel -r alice
```

```
sudo userdel -r bob
```

```
sudo groupdel developers
```

```
sudo groupdel testers
```

Verify:

id alice

id bob

getent group developers

getent group testers

No valid user or group records should remain.

Common Week 3 mistakes

Overwriting a file accidentally

command > file

replaces existing content. Use:

command >> file

to append.

Redirecting errors with only >

Incorrect:

ls /missing > error.txt

Correct:

ls /missing 2> error.txt

Forgetting to exit **tr**

Press:

Ctrl+D

Getting stuck in **vi**

Press:

Esc

:q!

to quit without saving.

Creating variables with spaces

Incorrect:

```
COURSE = SADS
```

Correct:

```
COURSE=SADS
```

Removing existing group memberships

Risky:

```
sudo usermod -G developers alice
```

Safer when adding a group:

```
sudo usermod -aG developers alice
```

Deleting a user but leaving the home directory

```
sudo userdel username
```

removes only the account. Use:

```
sudo userdel -r username
```

when the home directory should also be removed.

Week 3 essential commands

Command	Purpose
>	Redirect and overwrite standard output
>>	Redirect and append standard output
2>	Redirect standard error
<	Read standard input from a file
	Pipe output into another command
grep	Search matching lines

<code>sort</code>	Sort lines
<code>uniq</code>	Process adjacent duplicate lines
<code>cut</code>	Extract selected fields
<code>vi</code>	Edit text files
<code>export</code>	Convert a variable into an environment variable
<code>unset</code>	Remove a variable
<code>useradd</code>	Create a user
<code>usermod</code>	Modify a user
<code>userdel</code> <code>-r</code>	Delete a user and home directory
<code>groupadd</code>	Create a group
<code>groupdel</code>	Delete a group
<code>passwd</code>	Set or change a password
<code>chage</code>	Configure password ageing
<code>id</code>	Display UID, GID and group memberships
<code>getent</code>	Retrieve user or group database entries